

DISCIPLINE
EXECUTION
FREEDOM

PROCESS
PATIENCE
CONSISTENCY



ORDER FLOW SERIES

DELTA EXPLAINED

Footprint charts, delta, and absorption.

READ THE FLOW

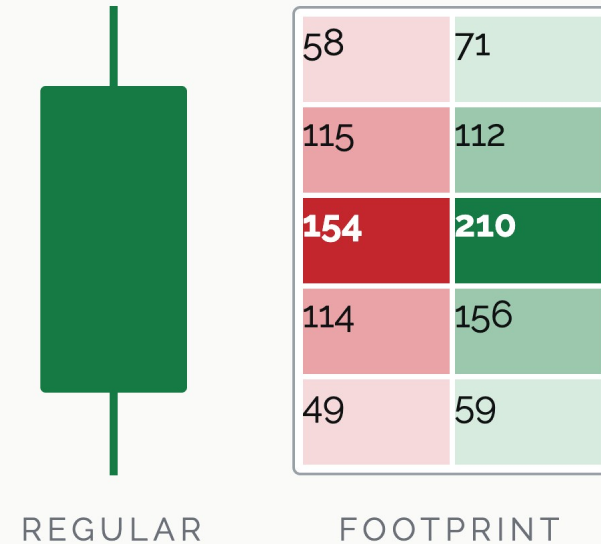
BLACKBRXXOR

TRADE YOUR EDGE

WHAT IS A FOOTPRINT CHART?

A footprint chart is a candlestick **opened up**. Instead of only open, high, low and close, it shows **how many contracts traded at every price** inside the candle.

Each price level is split in two: contracts sold into the **bid** on the left, contracts bought from the **ask** on the right.



REGULAR

FOOTPRINT

ANATOMY OF ONE CANDLE

PRICE	BID	ASK
21467.00	58	71
21466.00	115	112
21465.00	154	210
21464.00	114	156
21463.00	49	59

Each row = one price level

This chart groups 1.00 point (4 MNQ ticks) per row.

Left number = bid volume

Market sellers who hit the bid at that price.

Right number = ask volume

Market buyers who lifted the ask at that price.

BID SIDE VS ASK SIDE

LEFT SIDE

BID

- Aggressive selling
- Contracts that hit the bid
- Market sellers who wanted out now

RIGHT SIDE

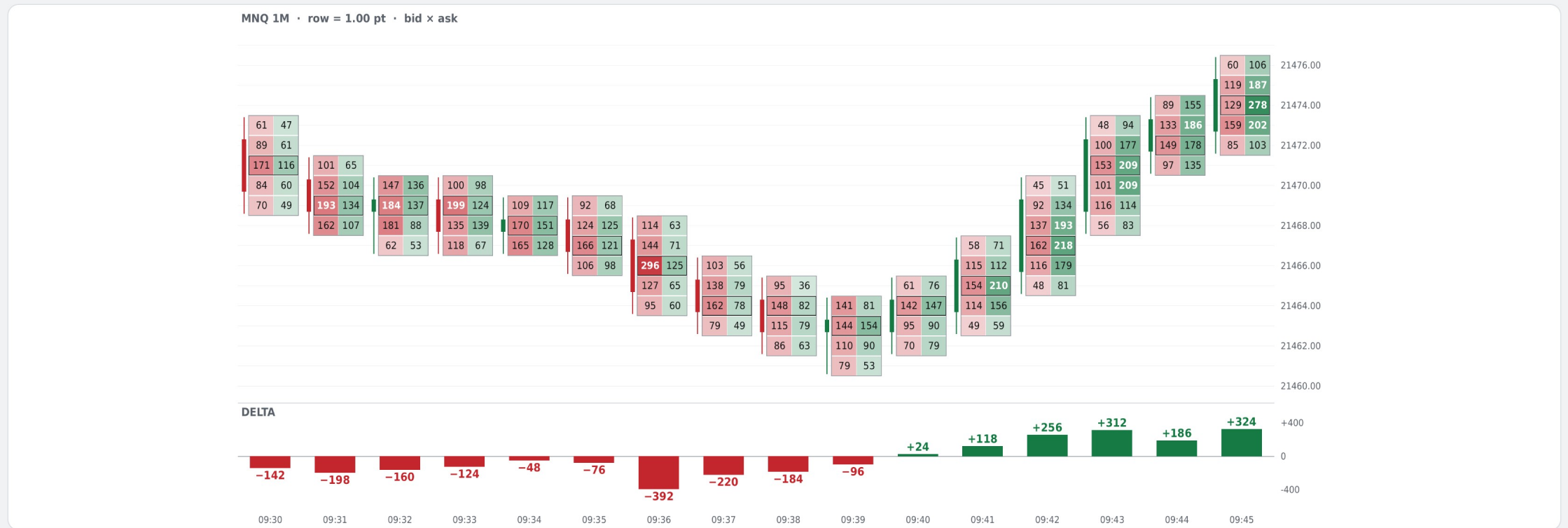
ASK

- Aggressive buying
- Contracts that lifted the ask
- Market buyers who wanted in now

"Aggressive" means a market order: the trader paid the spread to get filled immediately instead of waiting with a limit order.

A FULL SESSION

MNQ, 1-minute, 09:30 to 09:45. Every candle is a stack of bid × ask rows, with its delta underneath.



02 · WHAT IS DELTA?

$$\text{DELTA} = \text{ASK} - \text{BID}$$

The difference between aggressive buying and aggressive selling inside one candle.

WORKING OUT THE 09:41 CANDLE

TOTAL ASK

608

$71 + 112 + 210 + 156 + 59$

TOTAL BID

490

$58 + 115 + 154 + 114 + 49$

DELTA

+118

$608 - 490$

Total volume for the candle is $608 + 490 = \mathbf{1,098}$ contracts. Delta says buyers were the aggressors by 118 of them.

THREE WAYS DELTA PRINTS

NEGATIVE · 09:36

-392

392 more aggressive sells than aggressive buys.

Ask 384 – Bid 776. Sellers controlled the aggression.

NEAR ZERO · 09:40

+24

Aggressive buying and selling were about equal.

Ask 392 – Bid 368. Neither side had a clear edge.

POSITIVE · 09:43

+312

312 more aggressive buys than aggressive sells.

Ask 886 – Bid 574. Buyers controlled the aggression.

THE ORDER FLOW TABLE

Delta, total bid, total ask and volume for each minute. Every column adds up to its candle.

MNQ 1M · row = 1.00 pt · bid × ask



WHAT THE SESSION SAYS

09:30–09:39

Ten red bars in a row. Sellers are the aggressors, peaking at **-392** at 09:36, and price slides from 21473 to the 21461 low.

09:39–09:40

Selling fades to -96 at the low, then delta flips to +24. Aggression changes hands.

09:41–09:45

Five green bars, +118 to +324, while price climbs to a new high at 21476. Buyers are aggressive **and** price moves with them.

KEY TAKEAWAY

DELTA SHOWS WHO IS AGGRESSIVE, NOT WHO IS WINNING.

Always compare delta with price. Is that aggression actually moving the market?

04 · ABSORPTION

WHEN AGGRESSION HITS A WALL

What absorption is, how it shows up on the footprint, and how it changes the way you read delta.

WHAT IS ABSORPTION?

Aggressive market orders run into large **passive limit orders** at one price. The limit orders soak them up, so price **stops moving** even though delta is big.

THE AGGRESSOR

Market orders

Hits the bid or lifts the ask. **Shows up in delta.** This is the effort.

THE ABSORBER

Resting limit orders

Waits at a price and takes the other side. **Doesn't show up in delta.** You see it only as a wall.

DELTA IS EFFORT, PRICE IS RESULT

BIG DELTA · PRICE MOVES

INITIATIVE

Aggression is winning. The move is likely to continue.

BIG DELTA · PRICE STALLS

ABSORPTION

A passive wall is winning. Price turns, or the trend resumes.

SMALL DELTA · PRICE MOVES

THIN MARKET

Price moves with little opposition. Less conviction behind it.

SMALL DELTA · PRICE STALLS

BALANCE

Nobody is in control. Wait for one side to act.

AN ABSORPTION CANDLE

10:03

	BID	ASK
21480.00	96	342
21479.00	88	164
21478.00	72	118
21477.00	60	90
21476.00	44	62

Delta +416: huge buying effort

Ask 776 – Bid 360. The strongest buying of the move.

342 bought at 21480, none above

The heaviest print sits at the high, and 21481 never trades.

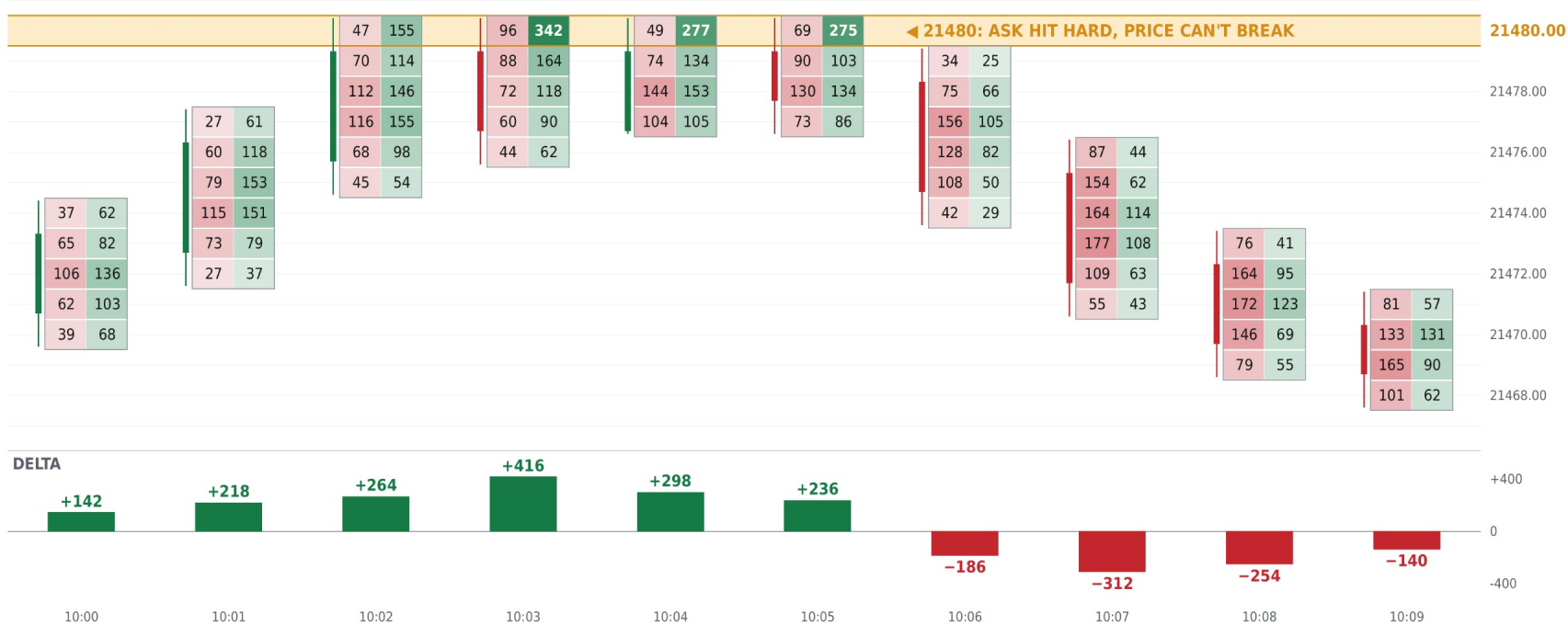
Closes at 21477, below its open

All that buying, and the candle still closes red.

SELL-SIDE ABSORPTION

Buyers pour into 21480 for four minutes. A passive seller holds it, and price drops once delta flips.

MNQ 1M · row = 1.00 pt · bid × ask · SELL-SIDE ABSORPTION



READING THE WALL AT 21480

10:02–10:05

Buyers keep lifting the ask: **+1,214** delta in four minutes and **1,049** contracts bought at 21480 alone. Price never prints 21481.

10:06

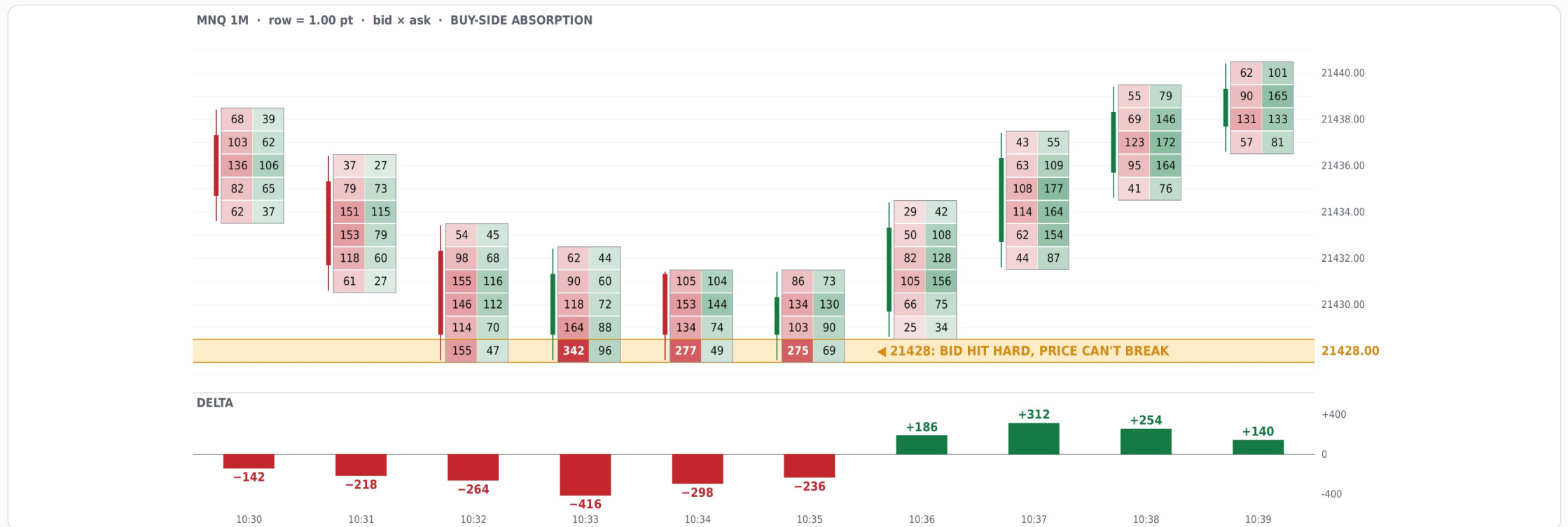
Delta flips to **-186**. Buyers stop pushing, and everyone who bought at 21480 is now underwater.

10:07–10:09

Sellers take over with **-706** delta, and price drops 12 points to 21468 as trapped buyers exit.

BUY-SIDE ABSORPTION

The mirror image at a low. Sellers pour into 21428, a passive buyer holds it, and price rallies once delta flips.



TWO KINDS OF ABSORPTION

TYPE 1

REVERSAL

The trend's **own** aggression gets absorbed at a high or low. The move is running out of fuel.

Next: the trend turns.

TYPE 2

CONTINUATION

The pullback's aggression **against** the trend gets absorbed. The trend's side is defending its level.

Next: the trend resumes.

Same footprint signature: big delta, no progress. **Context decides which one it is.**

CONTINUATION ABSORPTION

An uptrend pulls back. Sellers pour into 21448, a passive buyer holds it, and the trend resumes to new highs.



READING THE PULLBACK

11:03–11:05

The pullback. Sellers hit the bid, **-712** delta in three minutes, and **630** contracts sold into 21448. Price never prints 21447.

11:06

Delta is still **-198**, yet the candle closes green. Sellers are pushing and losing.

11:07–11:09

Buyers step back in with **+848** delta. Price breaks the 21454 high and runs to 21460.

TELLING THEM APART

	REVERSAL	CONTINUATION
Where	At the end of a move: a key high or low	On a pullback inside a trend
Who gets absorbed	The trend side: buyers at a high, sellers at a low	The counter-trend side: sellers in an uptrend, buyers in a downtrend
Delta	Strong with the trend, but price stalls	Strong against the trend, but the pullback stalls
What comes next	The trend turns	The trend resumes

CONFIRMED OR FAILED?

CONFIRMED WHEN

- Price rejects the level and closes back inside
- Delta flips to the other side
- The level holds on a retest
- Structure breaks the other way

FAILED WHEN

- Price trades and closes through the level
- Delta keeps pushing and price follows
- The passive order is used up
- Absorption turns into a breakout

Absorption is a warning, not an entry. Wait for the reaction.

KEY TAKEAWAY

**BIG DELTA + NO PROGRESS
= SOMEONE IS ABSORBING.**

At a high or low it warns of a reversal. On a pullback it signals continuation.
Trade the reaction, not the print.

05 · CVD

DELTA OVER TIME

Cumulative volume delta: the running score of who has been more aggressive all session.

WHAT IS CVD?

CVD = PREVIOUS CVD + THIS CANDLE'S DELTA

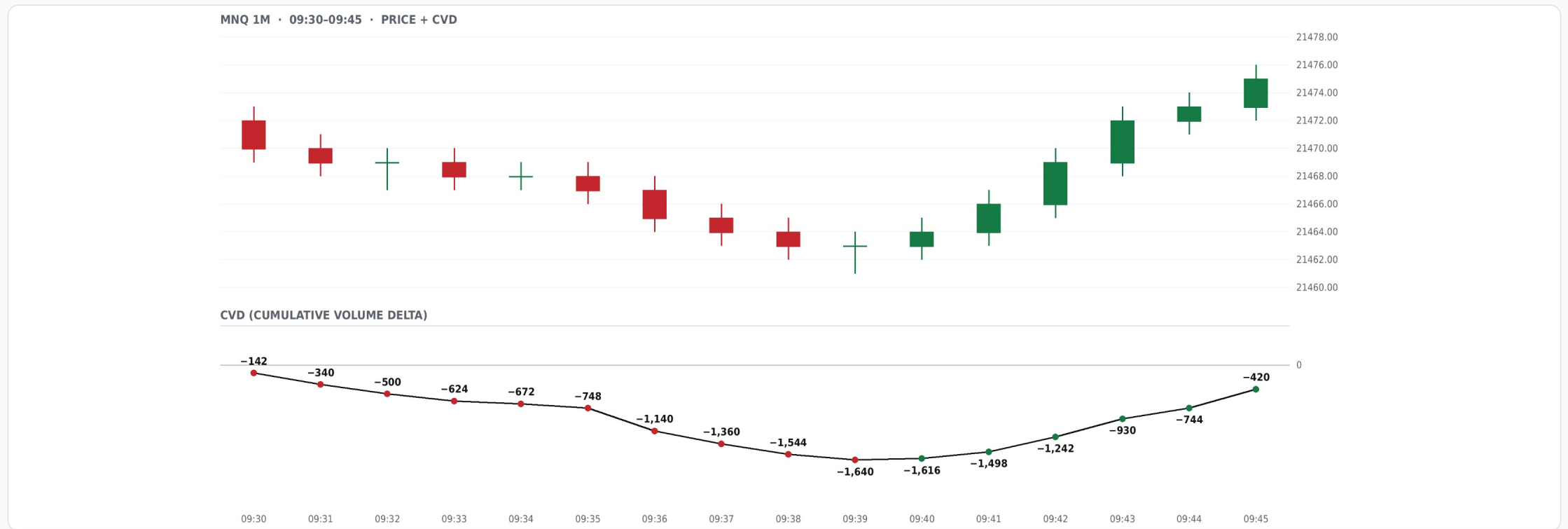
TIME	09:30	09:31	09:32	09:33	09:34	09:35
DELTA	-142	-198	-160	-124	-48	-76
CVD	-142	-340	-500	-624	-672	-748

Delta is one candle's score. CVD is the running total for the whole session.

05 · CVD

CVD ON THE 09:30 SESSION

The same sixteen candles as the delta chart, with delta added up into one line.



READ THE SLOPE, NOT THE SIGN

09:30–09:38

CVD falls every minute, from -142 to -1,544. Sellers are in control and price slides.

09:39

CVD bottoms at **-1,640** in the same minute price makes its 21461 low.

09:40–09:45

CVD climbs **1,220** to -420 while price makes a new high at 21476. Still below zero, but rising hard.

A rising line means buyers are in control right now, even if the total is still negative.

06 · DELTA DIVERGENCE

WHEN PRICE AND DELTA DISAGREE

Price makes a new extreme, but the aggression behind it is fading.

WHAT IS DELTA DIVERGENCE?

Price makes a **new high or low**, but CVD does **not** make a new high or low with it.

BEARISH

Price: higher high

CVD: lower high

Buyers pushed price up with less aggression than last time.

BULLISH

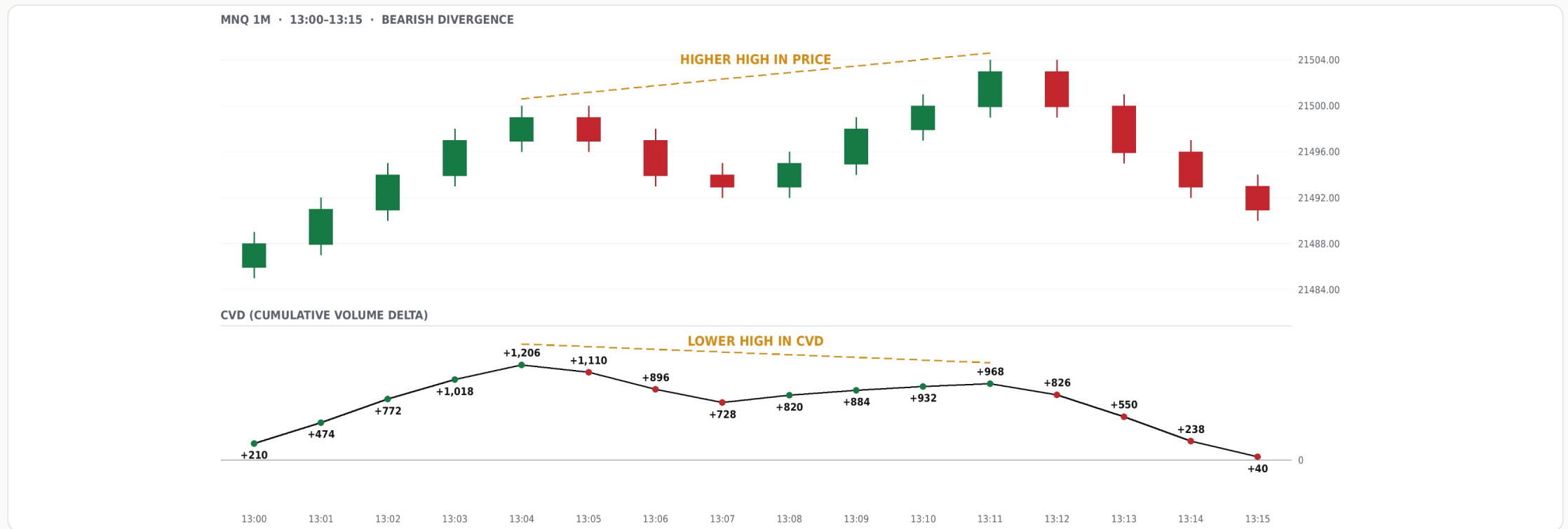
Price: lower low

CVD: higher low

Sellers pushed price down with less aggression than last time.

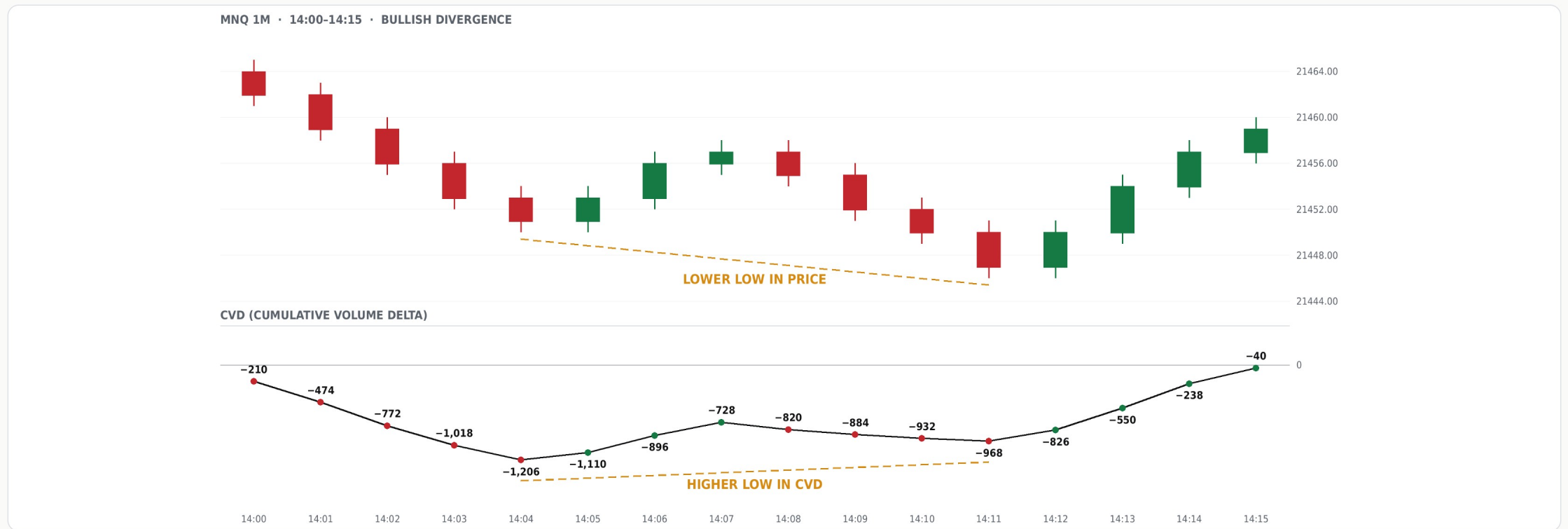
BEARISH DIVERGENCE

Price pushes to 21504, above the 21500 high, but CVD peaks lower. Then price rolls over.



BULLISH DIVERGENCE

Price drops to 21446, below the 21450 low, but CVD bottoms higher. Then price turns up.



COMPARING THE TWO PUSHES

BEARISH EXAMPLE	FIRST PUSH	SECOND PUSH
Time	13:00–13:04	13:08–13:11
Price high	21500	21504 · higher
Delta per candle	+188 to +298	+92 falling to +36
CVD at the high	+1,206	+968 · lower

A new high on weaker buying. The bullish example is the exact mirror at the lows.

DIVERGENCE VS ABSORPTION

ABSORPTION

Effort without result

Big delta, but price is stuck at one level. A passive wall is taking the hits.

DIVERGENCE

Result without effort

Price reaches a new extreme, but with weaker delta. The push is running on fumes.

Both say the same thing: **the aggressive side is losing its grip**. When they show up together, pay attention.

A WARNING, NOT A TRIGGER

STRONGER WHEN

- It forms at a key high or low
- Absorption shows up at the extreme
- Delta flips to the other side
- Structure breaks the other way

WEAKER WHEN

- The bigger trend is strong
- The move is news-driven
- The CVD difference is small
- Nothing has confirmed it yet

Divergence can stack up several times in a strong trend. Wait for price to confirm it.

WHAT'S BEHIND THE WALL

Large hidden orders that keep refilling at the same price.

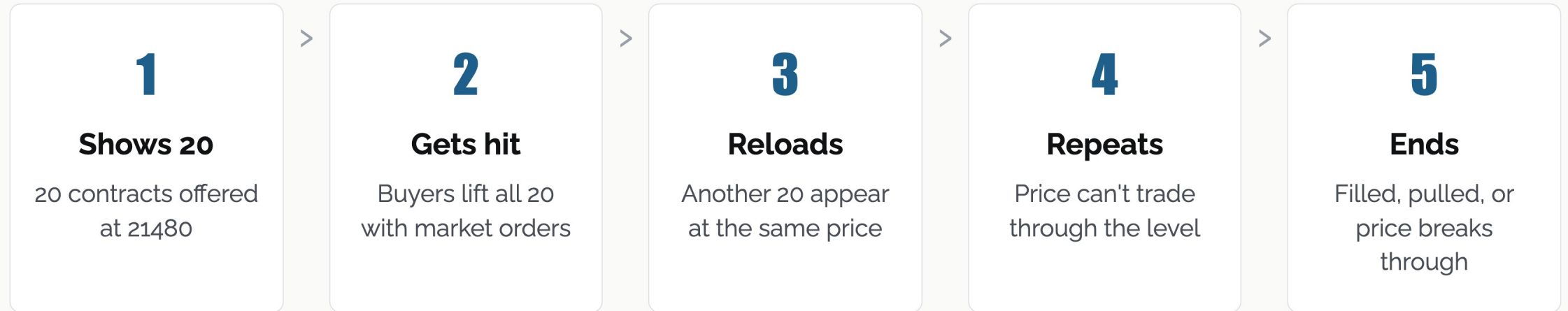
WHAT IS AN ICEBERG ORDER?

A **large limit order** that shows only a **small piece** in the order book.

When the visible piece fills, the next piece reloads at the same price. The true size stays hidden, like the part of an iceberg under the water.



HOW IT REFILLS



The book only ever shows 20. The trades at that price keep adding up.

HOW TO SPOT ONE

TRADED AT 21480

1,049

—

SHOWN IN THE BOOK

20

=

HIDDEN, AT LEAST

1,029

Far more traded at a price than was ever displayed there

The level reloads right after every hit

Heavy prints at the same price, candle after candle

Price can't move through it, even with big delta

SELL-SIDE VS BUY-SIDE

ACTS AS RESISTANCE

SELL-SIDE ICEBERG

A hidden sell order above price soaks up aggressive buyers.

Footprint: heavy **ask** prints at one price, positive delta, price stalls.

EXAMPLE: THE 21480 WALL

ACTS AS SUPPORT

BUY-SIDE ICEBERG

A hidden buy order below price soaks up aggressive sellers.

Footprint: heavy **bid** prints at one price, negative delta, price holds.

EXAMPLES: 21428 AND 21448

LIMITATIONS

Not every big print is an iceberg

It can be one large order, or a burst on news.

It can be pulled at any time

The wall is only there while its owner wants it.

It doesn't always mean reversal

It may just be someone building or exiting a position.

Once it's used up, price can run

A broken iceberg level can turn into a fast breakout.

Confirm with CVD, structure and context before acting on it.

KEY TAKEAWAY

**THE BOOK SHOWS 20.
THE TAPE SHOWS 1,049.**

An iceberg is hidden size defending a price. It's often what's behind absorption.



RECAP

THE FULL PICTURE

FOOTPRINT

Bid × ask volume at every price

DELTA

Ask minus bid, one candle at a time

ABSORPTION

Big delta, no progress

CVD

Delta added up over the session

DIVERGENCE

New price extreme, weaker CVD

ICEBERGS

Hidden size behind the wall

DISCIPLINE

EXECUTION

FREEDOM